

10.7.105

EE25BTECH11057 - Rushil Shanmukha Srinivas

Question: Consider the parabola $y^2 = 4x$. Let **S** be the focus of the parabola. A pair of tangents drawn to the parabola from the point **R** = (-2,1) meet the parabola at **R₁** and **R₂**. Let **Q₁** and **Q₂** be points on the lines **SR₁** and **SR₂** respectively such that **RQ₁** is perpendicular to **SR₁** and **RQ₂** is perpendicular to **SR₂**. Then, which of the following is/are TRUE ?

$$1) SQ_1 = 2$$

$$3) RQ_1 = 3$$

$$2) Q_1Q_2 = \frac{3\sqrt{10}}{5}$$

$$4) SQ_2 = 1$$

Solution:

Name	Value
Parabola	$\mathbf{x}^T \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x} - (4 \ 0) \mathbf{x} = 0$

Table : Conics

On comparing Parabola with

$$\mathbf{x}^T \mathbf{V} \mathbf{x} + 2\mathbf{u}^T \mathbf{x} + f = 0 \quad (4.1)$$

The parameters of the Parabola are

$$\mathbf{V} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \mathbf{u} = \begin{pmatrix} -2 \\ 0 \end{pmatrix}, f = 0 \quad (4.2)$$

As **V** is an upper triangular matrix, we get the eigen values as the diagonal entries

$$\lambda_1 = 0 \quad \lambda_2 = 1 \quad (4.3)$$

For a Symmetric Matrix, we have the Eigen decomposition

$$\mathbf{V} = \mathbf{P} \mathbf{D} \mathbf{P}^T \quad (4.4)$$

$$\mathbf{D} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{V} \quad (4.5)$$

$$\mathbf{P} = (\mathbf{p}_1 \ \mathbf{p}_2) \text{ and } \mathbf{P}^T \mathbf{P} = \mathbf{I} \quad (4.6)$$

Solving both

$$\mathbf{V} = \mathbf{P} \mathbf{V} \mathbf{P}^T \text{ and } \mathbf{P}^T \mathbf{P} = \mathbf{I} \quad (4.7)$$

We get

$$\mathbf{P} = \begin{pmatrix} \pm 1 & 0 \\ 0 & \pm 1 \end{pmatrix} \quad (4.8)$$

So

$$\mathbf{p}_1 = \begin{pmatrix} \pm 1 \\ 0 \end{pmatrix} \text{ and } \mathbf{p}_2 = \begin{pmatrix} 0 \\ \pm 1 \end{pmatrix} \quad (4.9)$$

For a parabola, Eccentricity = $e = 1$

$$\mathbf{n} = \sqrt{\lambda_2} \mathbf{p}_1 = \mathbf{p}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (4.10)$$

$$c = \frac{\|\mathbf{u}\|^2 - \lambda_2 f}{2\mathbf{u}^\top \mathbf{n}} = \frac{4}{-4} = -1 \quad (4.11)$$

$$Focus = \mathbf{F} = \mathbf{S} = \frac{ce^2 \mathbf{n} - \mathbf{u}}{\lambda_2} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (4.12)$$

Given,

$$\mathbf{R} = \begin{pmatrix} -2 \\ 1 \end{pmatrix} \quad (4.13)$$

Given, The point of contact $\mathbf{q}$ the equation of tangent to (4.1) is

$$(\mathbf{V}\mathbf{q} + \mathbf{u})^\top \mathbf{x} + \mathbf{u}^\top \mathbf{q} + f = 0 \quad (4.14)$$

As the tangent passes through $\mathbf{R}$

$$(\mathbf{V}\mathbf{q} + \mathbf{u})^\top \mathbf{R} + \mathbf{u}^\top \mathbf{q} + f = 0 \quad (4.15)$$

Also $\mathbf{q}$ lies on the conic

$$\mathbf{q}^\top \mathbf{V}\mathbf{q} + 2\mathbf{u}^\top \mathbf{q} = 0 \quad (4.16)$$

$$\mathbf{V}^\top = \mathbf{V}, (\mathbf{V}\mathbf{q} + \mathbf{u})^\top \mathbf{R} + \mathbf{u}^\top \mathbf{q} = \mathbf{q}^\top \mathbf{V}\mathbf{R} + \mathbf{u}^\top \mathbf{R} + \mathbf{u}^\top \mathbf{q} = 0 \quad (4.17)$$

$$\mathbf{V}\mathbf{R} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{ and } \mathbf{V}\mathbf{R} + \mathbf{u} = \mathbf{R} \text{ and } \mathbf{u}^\top \mathbf{R} = 4 \quad (4.18)$$

Also $\mathbf{q}^\top \mathbf{V}\mathbf{R}, \mathbf{R}^\top \mathbf{V}\mathbf{q}$ are scalars. For a scalar the transpose of a scalar is the same scalar. So

$$\mathbf{q}^\top \mathbf{V}\mathbf{R} = (\mathbf{q}^\top \mathbf{V}\mathbf{R})^\top = \mathbf{R}^\top \mathbf{V}\mathbf{q} \quad (4.19)$$

From (4.17)

$$\mathbf{q}^\top \mathbf{V}\mathbf{R} + 4 + (\mathbf{R} - \mathbf{V}\mathbf{R})^\top \mathbf{q} = 0 \quad (4.20)$$

$$\mathbf{q}^\top \mathbf{V}\mathbf{R} + 4 + \mathbf{R}^\top \mathbf{q} - \mathbf{R}^\top \mathbf{V}\mathbf{q} = 0 \quad (4.21)$$

$$\mathbf{R}^\top \mathbf{q} = -4 \quad (4.22)$$

$\mathbf{q}$ can be expressed as combination of $\mathbf{R}$ and its orthogonal component

$$\mathbf{q} = \lambda \mathbf{R} + \mathbf{z}, \text{ where } \mathbf{R}^\top \mathbf{z} = 0 \quad (4.23)$$

$$\mathbf{R}^\top \mathbf{q} = \lambda \mathbf{R}^\top \mathbf{R} = -4 \quad (4.24)$$

$$\lambda = \frac{-4}{5} \quad (4.25)$$

Substituting (4.23) in (4.16)

$$\lambda^2 \mathbf{R}^\top \mathbf{V} \mathbf{R} + 2\lambda \mathbf{R}^\top \mathbf{V} \mathbf{z} + \mathbf{z}^\top \mathbf{V} \mathbf{z} + 8\lambda + 2\mathbf{u}^\top \mathbf{z} = 0 \quad (4.26)$$

$$\mathbf{R}^\top \mathbf{V} \mathbf{R} = 1 \text{ and } \lambda = \frac{-4}{5} \quad (4.27)$$

Substituting these and simplifying the equation

$$25\mathbf{z}^\top \mathbf{V} \mathbf{z} - 40\mathbf{R}^\top \mathbf{V} \mathbf{z} + 50\mathbf{u}^\top \mathbf{z} - 144 = 0 \quad (4.28)$$

$$\mathbf{z} = \begin{pmatrix} \frac{-\mathbf{u}^\top \mathbf{z}}{\mathbf{R}^\top \mathbf{V} \mathbf{z}} \end{pmatrix} \quad (4.29)$$

$$\mathbf{R}^\top \mathbf{z} = 0 \implies \mathbf{R}^\top \mathbf{V} \mathbf{z} = -\mathbf{u}^\top \mathbf{z} \quad (4.30)$$

Let

$$\mathbf{R}^\top \mathbf{V} \mathbf{z} = -\mathbf{u}^\top \mathbf{z} = k \quad (4.31)$$

Then

$$\mathbf{z}^\top \mathbf{V} \mathbf{z} = \left(\mathbf{R}^\top \mathbf{V} \mathbf{z} \right)^2 = k^2 \quad (4.32)$$

Then from (4.28)

$$25k^2 - 90k - 144 = 0 \quad (4.33)$$

$$\left(k - \frac{24}{5} \right) \left(k + \frac{6}{5} \right) = 0 \quad (4.34)$$

$$k = \frac{24}{5} \text{ or } k = \frac{-6}{5} \quad (4.35)$$

$$\mathbf{z} = \frac{1}{5} \begin{pmatrix} 12 \\ 24 \end{pmatrix} \text{ or } \mathbf{z} = \frac{1}{5} \begin{pmatrix} -3 \\ -6 \end{pmatrix} \quad (4.36)$$

$$\mathbf{q} = \begin{pmatrix} 4 \\ 4 \end{pmatrix} \text{ or } \mathbf{q} = \begin{pmatrix} 1 \\ -2 \end{pmatrix} \quad (4.37)$$

As $\mathbf{R}_1, \mathbf{R}_2$ are the point of contact of Tangents $\{\mathbf{R}_1, \mathbf{R}_2\} \in \mathbf{q}$

$$\mathbf{R}_1 = \begin{pmatrix} 4 \\ 4 \end{pmatrix} \text{ and } \mathbf{R}_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix} \quad (4.38)$$

If $\mathbf{A}, \mathbf{B}$ are on line $\mathbf{h}^\top \mathbf{x} = d$ then

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \end{pmatrix}^\top \mathbf{h} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (4.39)$$

For the line passing through $\mathbf{R}_1$ and $\mathbf{S}$

$$\begin{pmatrix} 4 & 4 \\ 1 & 0 \end{pmatrix} \mathbf{h} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (4.40)$$

The augmented Matrix is

$$\left(\begin{array}{cc|c} 4 & 4 & 1 \\ 1 & 0 & 1 \end{array} \right) \xleftrightarrow{R_2 \rightarrow R_2 - \frac{R_1}{4}} \left(\begin{array}{cc|c} 4 & 4 & 1 \\ 0 & -1 & \frac{3}{4} \end{array} \right) \xleftrightarrow{R_1 \rightarrow R_1 + 4R_2} \left(\begin{array}{cc|c} 4 & 0 & 4 \\ 0 & -1 & \frac{3}{4} \end{array} \right) \quad (4.41)$$

$$\left(\begin{array}{cc|c} 4 & 0 & 4 \\ 0 & -1 & \frac{3}{4} \end{array} \right) \xleftrightarrow{R_1 \rightarrow -\frac{R_1}{4}} \left(\begin{array}{cc|c} -1 & 0 & -1 \\ 0 & -1 & \frac{3}{4} \end{array} \right) \quad (4.42)$$

$$\mathbf{h} = \begin{pmatrix} 1 \\ -\frac{3}{4} \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 4 \\ -3 \end{pmatrix}, \mathbf{m} = \begin{pmatrix} 3 \\ 4 \end{pmatrix} \quad (4.43)$$

The equation of the line through $\mathbf{R}_1, \mathbf{S}$ is

$$(4 \quad -3) \mathbf{x} = 4 \quad (4.44)$$

For the line passing through $\mathbf{R}_2$ and $\mathbf{S}$

$$\begin{pmatrix} 1 & -2 \\ 1 & 0 \end{pmatrix} \mathbf{h} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (4.45)$$

The augmented Matrix is

$$\left(\begin{array}{cc|c} 1 & -2 & 1 \\ 1 & 0 & 1 \end{array} \right) \xleftrightarrow{R_2 \rightarrow R_2 - R_1} \left(\begin{array}{cc|c} 1 & -2 & 1 \\ 0 & 2 & 0 \end{array} \right) \xleftrightarrow{R_1 \rightarrow R_1 + R_2} \left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 2 & 0 \end{array} \right) \quad (4.46)$$

$$\left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 2 & 0 \end{array} \right) \xleftrightarrow{R_2 \rightarrow \frac{R_2}{2}} \left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 0 \end{array} \right) \quad (4.47)$$

$$\mathbf{h} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \mathbf{m} = \begin{pmatrix} 0 \\ 2 \end{pmatrix} \quad (4.48)$$

The equation of the line through $\mathbf{R}_2, \mathbf{S}$ is

$$(1 \quad 0) \mathbf{x} = 1 \quad (4.49)$$

If $\mathbf{Q}$ be the foot of perpendicular from $\mathbf{T}$ to the line $\mathbf{h}^\top \mathbf{x} = d$ then

$$\begin{pmatrix} \mathbf{m} & \mathbf{h} \end{pmatrix}^\top \mathbf{Q} = \begin{pmatrix} \mathbf{m}^\top \mathbf{T} \\ d \end{pmatrix} \quad (4.50)$$

$\mathbf{Q}_1, \mathbf{Q}_2$ are foot of perpendicular drawn from $\mathbf{R}$ to (4.44) and (4.49). So

$$\begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix} \mathbf{Q}_1 = \begin{pmatrix} -2 \\ 4 \end{pmatrix} \quad (4.51)$$

$$\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} \mathbf{Q}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad (4.52)$$

The augmented Matrix for (4.51) is

$$\left(\begin{array}{cc|c} 3 & 4 & -2 \\ 4 & -3 & 4 \end{array} \right) \xleftrightarrow{R_2 \rightarrow 3R_2 - 4R_1} \left(\begin{array}{cc|c} 3 & 4 & -2 \\ 0 & -25 & 20 \end{array} \right) \xleftrightarrow{R_1 \rightarrow 25R_1 + 4R_2} \left(\begin{array}{cc|c} 75 & 0 & 30 \\ 0 & -25 & 20 \end{array} \right) \quad (4.53)$$

$$\left(\begin{array}{cc|c} 75 & 0 & 30 \\ 0 & -25 & 20 \end{array} \right) \xleftrightarrow{R_2 \rightarrow \frac{-R_2}{25}} \left(\begin{array}{cc|c} 75 & 0 & 30 \\ 0 & 1 & \frac{-4}{5} \end{array} \right) \xleftrightarrow{R_1 \rightarrow \frac{R_1}{75}} \left(\begin{array}{cc|c} 1 & 0 & \frac{2}{5} \\ 0 & 1 & \frac{-4}{5} \end{array} \right) \quad (4.54)$$

$$\mathbf{Q}_1 = \begin{pmatrix} \frac{2}{5} \\ \frac{-4}{5} \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 2 \\ -4 \end{pmatrix} \quad (4.55)$$

The augmented Matrix for (4.52) is

$$\left(\begin{array}{cc|c} 0 & 2 & 2 \\ 1 & 0 & 1 \end{array} \right) \xleftrightarrow{R_1 \rightarrow \frac{R_1}{2}} \left(\begin{array}{cc|c} 0 & 1 & 1 \\ 1 & 0 & 1 \end{array} \right) \xleftrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \end{array} \right) \quad (4.56)$$

$$\mathbf{Q}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (4.57)$$

$$SQ_1 = \|\mathbf{S} - \mathbf{Q}_1\| = \left\| \frac{1}{5} \begin{pmatrix} 3 \\ 4 \end{pmatrix} \right\| = 1 \quad (4.58)$$

$$Q_1Q_2 = \|\mathbf{Q}_1 - \mathbf{Q}_2\| = \left\| \frac{1}{5} \begin{pmatrix} 3 \\ 9 \end{pmatrix} \right\| = \frac{3\sqrt{10}}{5} \quad (4.59)$$

$$RQ_1 = \|\mathbf{R} - \mathbf{Q}_1\| = \left\| \frac{1}{5} \begin{pmatrix} -12 \\ 9 \end{pmatrix} \right\| = 3 \quad (4.60)$$

$$SQ_2 = \|\mathbf{S} - \mathbf{Q}_2\| = \left\| \begin{pmatrix} 0 \\ -1 \end{pmatrix} \right\| = 1 \quad (4.61)$$

So options 2,3,4 are correct.

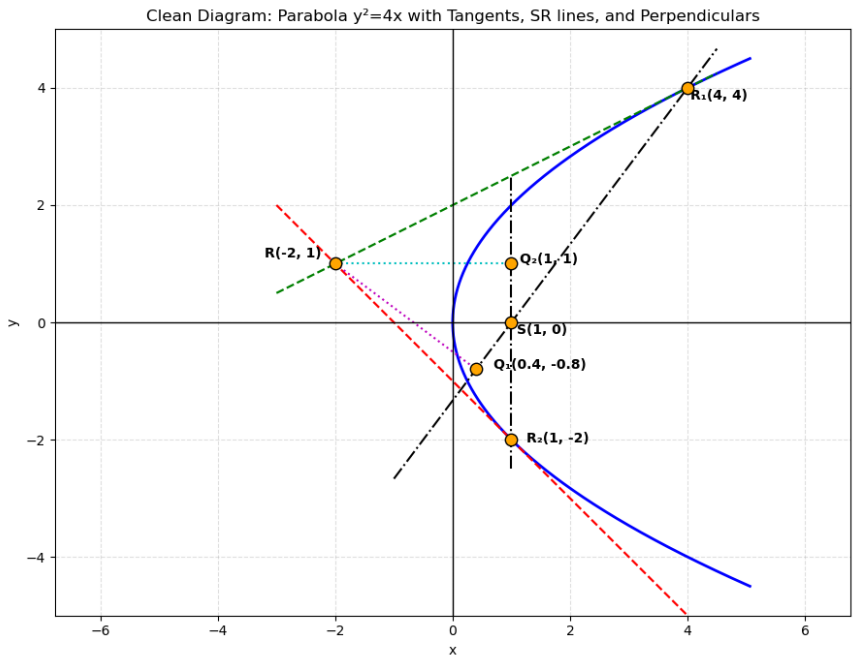


Fig : Parabola and Tangents